

FINAL EXPLANATION: BIOLOGY GRADE 11

Student Assessment Document

FINAL EXPLANATION: BIOLOGY GRADE 11 – SUB-STRAND 1.4 CELL DIVISION: FINAL EXPLANATION

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

Use this model to help you write your own Final Explanation answering our driving question: 'How does the body produce the enormous number of new cells needed for growth?' Think about the baby-to-teenager phenomenon: a newborn of about 3 kg becomes a teenager of 50 kg or more. Your explanation should (1) name the process that makes new cells, (2) use evidence from our investigations (the onion root tip or model of mitosis, the chromosome-copying model, and the meiosis model), (3) explain how each new cell gets the right instructions, and (4) end with a clear conclusion. Write in your own words, use correct scientific terms, and use Kenyan examples where they help. Compare the exemplar with your draft, then revise.

1. Claim: Where do the new cells come from?

State your claim in one or two sentences. What process produces the billions of new cells a child needs to grow from a 3 kg baby to a 50 kg teenager? Remember, we only eat food (like ugali, sukuma wiki and githeri), so where do the cells come from?

The body produces new cells through cell division. You do not get new cells from food directly. Food such as ugali, beans and sukuma wiki gives you the energy and building materials, but your existing cells use them to divide and make more cells. A single fertilised cell divides again and again, and this is how a newborn of about 3 kg grows into a teenager of 50 kg or more.

2. Evidence: What did we observe about mitosis?

Use evidence from our investigations to explain how one cell becomes two. Describe what happens to the chromosomes during mitosis and why the new cells receive

In our investigations, we observed that one parent cell can divide to make new cells, which shows that cell division increases cell number. When we modelled mitosis, we saw that the chromosomes are copied first, then lined up and pulled apart so each end of the cell gets a full set. The cell then splits into two daughter cells. Because the chromosomes were copied exactly, both daughter cells are genetically similar to the parent cell and carry the same instructions (DNA). This is why a new skin, bone or muscle cell can do its job properly. Mitosis repeated many times gives the enormous number of cells needed for growth, development, repair and replacement, for example when a wound on your knee heals after a fall while playing football.

the correct instructions. Mention what you saw or modelled.	
---	--

3. Evidence: What is the role of meiosis?

Explain how meiosis is different from mitosis. How many cells are made, what is their chromosome number, and why does this matter for reproduction? Use evidence from our meiosis model.	In our meiosis model, we observed one cell producing four cells, each with half the original chromosome number. Mitosis makes two cells with the full set of chromosomes for growth and repair, but meiosis makes reproductive cells (gametes: sperm and egg cells) for sexual reproduction. Having half the chromosomes is important because when a sperm cell fertilises an egg cell, the full chromosome number is restored in the zygote. Without meiosis, the chromosome number would double in every generation. The zygote then grows by mitosis into a baby, which links reproduction back to growth.
---	---

4. Other ways organisms make new cells

Give examples of how some other organisms reproduce using cell division. Name at least two methods and say how they are similar to or different from sexual reproduction.	Some organisms reproduce without needing two parents. Bacteria reproduce by binary fission, where one cell copies its DNA and splits into two new cells. Yeast, used to make mandazi dough rise, reproduces by budding, where a small bud grows on the parent cell and then separates. These methods use cell division to make offspring that are very similar to the parent, unlike sexual reproduction, which uses gametes from meiosis and produces offspring that differ from both parents.
--	---

5. Conclusion: Putting it all together

Write a short conclusion that answers the driving question and links growth, development, repair and reproduction. Connect back to the child growing from infancy to adolescence.	Therefore, cell division is essential for the growth, development, repair and reproduction of living organisms. Mitosis produces genetically similar daughter cells that increase the number of body cells as a child grows, and it also replaces and repairs damaged cells. Meiosis produces reproductive cells with half the chromosome number needed for sexual reproduction, and some organisms reproduce by budding or binary fission. So the billions of new cells in a growing child come from existing cells dividing, with each new cell receiving a copy of the instructions it needs.
--	--

RUBRIC

Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
Claim and use of scientific ideas	Clearly states that new cells are made by cell division from existing cells and explains that food supplies energy and materials, not cells. Uses all key terms	States that cell division makes new cells and uses most key terms correctly, with minor errors or missing links to food or growth.	Claim is unclear or incomplete, or key terms are missing or used incorrectly.

	(mitosis, meiosis, chromosomes, daughter cells, gametes) correctly.		
Explanation of mitosis and how cells get the right instructions	Accurately explains that chromosomes are copied and shared equally so two genetically similar daughter cells form, and links this to growth, development, repair and replacement with a Kenyan example.	Describes mitosis producing two similar cells and links it to growth or repair, but with limited detail about chromosome copying or examples.	Description of mitosis is vague or inaccurate, with little or no mention of chromosomes or genetic information.
Explanation of meiosis and other forms of reproduction	Correctly explains that meiosis makes four cells with half the chromosome number, why this is needed for fertilisation, and gives accurate examples of binary fission and budding.	Explains meiosis and gives at least one other method of reproduction, with minor gaps or inaccuracies.	Confuses meiosis with mitosis or gives little or no information about other types of reproduction.
Use of evidence from investigations	Refers specifically to observations from our investigations (models, microscope or images) and uses them to support each part of the explanation.	Refers to some investigation evidence, but links to claims are general.	Little or no evidence from investigations is used.
Conclusion and communication	Gives a clear, logical conclusion linking growth, development, repair and reproduction back to the phenomenon of a child growing; writing is well organised and easy to follow.	Gives a conclusion that answers the driving question, with some organisation issues.	Conclusion is missing or does not answer the driving question; writing is hard to follow.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.